ITEM 1

S/N	BASIS	BASIS CODE	LEVEL	DESCRIPTOR AND SUCCESS CRITERIA	WEIGHT
1	AIM	A	A(2)	States relevant aim; Correct statement with quantity and Purpose. Eg. An experiment to determine the force constant of the spring provided and advise for their use for making spring pens.	5
			A(1)	States relevant aim; Partial statement with quantity only.	2
			A(0)	States Irrelevant aim or purpose only.	0
2	VARIABLES	VAR	VAR(3)	Presents at least 4 variables (4–6) Eg. Independent – mass ' mass / force Dependent – new position / extension caused Controlled – initial pointer position	5
			VAR(2)	2–3 Variables	4
			VAR(1)	1 Variable	3
			VAR(0)	States Irrelevant variables	0
3	HYPOTHESIS	H	H(1)	States relevant statement with quantity, realistic range of value, geared towards addressing the problem Eg: The force constant of the spring ranges from 18Nm^{-1} to 25 Nm^{-1}.	5
			H(0)	States irrelevant Statement	0

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4	**PROCEDURE**	**P**	**P(3)**	• List of the relevant apparatus used in the experiment. • Labelled illustration of working experimental set up. • Coherent, necessary steps written in past tense. • Including statement about at least 5 realistic values of the independent variable. • Range 8–12 key points, working set up is a must.	**6**
			P(2)	• Range 4–7 key points, working set up is a must	**4**
			P(1)	• Range 1–3 key points with or without a set up	**3**
			P(0)	• No / Irrelevant procedure	**0**
5	**ERRORS AND PRECAUTIONS**	**Ep**	**Ep(2)**	States two relevant errors and their corresponding Precautions.	**5**
			Ep(1)	States one relevant error and its Corresponding Precaution.	**3**
			Ep(0)	No / Irrelevant error or error only.	**0**
6	**DATA PRESENTATION**	**PD**	**PD(3)**	• Realistic organised data of corresponding independent and dependent variables in a suitable table. • Appropriate units of independent and dependent variables labelled in appropriate columns. • At least 5 sets of independent and dependent variable values. • Values recorded in expected trend. **Note:** Values of new Pointer position must first be recorded in **cm** to **1 d.p**	**6**

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			PD(2)	• Realistic organised data of corresponding independent and dependent variables in a suitable table. • Appropriate units of independent and dependent variables labelled in appropriate columns. • 3–4 sets of independent and dependent variable values • Values recorded in expected trend.	**4**
			PD(1)	• Realistic organised data of corresponding independent and dependent variables in a suitable table. • Appropriate units of independent and dependent variables labelled in appropriate columns. • 1–2 sets of independent and dependent variable values • Values recorded in expected trend.	**2**
			PD(0)	• No set of data	**0**
7	**RECORDING DATA WITH ACCURACY**	**RD**	**RD(2)**	• At least three (3) dependent variable data recorded within the expected range • At least 3 values of dependent variable data recorded to appropriate decimal places in accordance with the instrument used. • Correct significant figures or decimal places for calculated values!.	**4**
			RD(1)	• 1–2 dependent variable data recorded within expected range. • Dependent variable data has appropriate decimal places in accordance with the instrument used.	**3**

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				• Correct significant figures or decimal places for calculated values.	
			RD(0)	• No / Irrelevant data recorded	**0**
8	**DATA ANALYSIS**	**DA**	**DA(3)**	• Correct title of the graph • Vertical and horizontal axes with starting points clearly indicated. • Labelling of the axes with quantity and unit. • Suitable and uniform Scales on vertical and horizontal axis. • Correct plotting of at least 5 points. • Line of best fit balancing all the correctly plotted points.	**5**
			DA(2)	• Correct title of the graph • Vertical and horizontal axes with starting points clearly indicated. • Labelling of the axes with quantity and unit. • Suitable and uniform Scales on vertical and horizontal axis. • Correct plotting of 3- 4 points. • Line of best fit balancing all the correctly plotted points.	**4**
			DA(1)	• Correct title of the graph • Vertical and horizontal axes with starting points clearly indicated. • Labelling of the axes with quantity and unit.	**3**

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				• Suitable and uniform Scales on vertical and horizontal axis. • Correct plotting of 1–2 points. • Line of best fit balancing all the correctly plotted points.	
			DA(0)	• No point plotted correctly	**0**
9	**DATA INTERPRENTATION**	**DI**	**DI(2)**	• Substitutes all the four values correctly in the slope formula. • Obtains correct value based on the substituted values. • Correct unit of the value of slope. • Comparison of the value obtained with the range values.	**5**
			DI(1)	• Substitutes 2–3 values correctly in the slope values formula. • Obtains correct value based on the substituted values. • Correct unit of the value of slope. • Comparison of the value obtained with the range values	**4**
			DI(0)	• Less than 2 values substituted in the slope formula.	**0**
10	**ADVICE OR DECISION**	**CAd**	**CAd(1)**	• Based on the obtained value and its position in the expected, gives advice or decision for next course of action i.e Conclusion with correct advice.	**6**
			CAd(0)	Conclusion without correct advice	**0**

Total Maximum Weight = 52

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ITEM 2

S/N	BASIS	BASIS CODE	LEVEL	DESCRIPTOR	WEIGHT
1	**AIM**	**A**	A(2)	States relevant aim; Correct statement with quantity and Purpose. Eg. An experiment to determine the **resistance** of 6cm of the conducting material provided and advise for its use in **repairing the extension cable plug.**	**5**
			A(1)	States relevant aim; Partial statement with quantity only.	**2**
			A(0)	States Irrelevant aim or purpose only.	**0**
2	**VARIABLES**	VAR	VAR(3)	Presents at least 4 variables (4–6) Eg. Independent – the current through the conductor. Dependent – the P.d across the conductor Controlled – the emf of the driver cell, temperature diameter, length of conducting material	**5**
			VAR(2)	2–3 Variables	**4**
			VAR(1)	1 Variable	**3**
			VAR(0)	States Irrelevant variables	**0**
3	**HYPOTHESIS**	**H**	H(1	States relevant statement with quantity, realistic range of value, geared towards addressing the problem Eg: The resistance of the conducting material ranges from 0.2Ω to 1.2Ω	**5**
			H(0	States irrelevant Statement	**0**

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4	**PROCEDURE**	**P**	P(3)	• List of the relevant apparatus used in the experiment. • Labelled illustration of working experimental set up. • Coherent, necessary steps written in past tense. • Including statement about at least 5 realistic values of the independent variable. • Range 7–10 key points, working circuit is a must.	**6**
			P(2)	• Range 3–6 key points, working set up is a must	**4**
			P(1)	• Range 1–2 key points with or without a set up	**3**
			P(0)	• No / Irrelevant procedure	**0**
5	**ERRORS AND PRECAUTIONS**	**Ep**	Ep(2	States two relevant errors and their corresponding Precautions.	**5**
			Ep(1	States one relevant error and its Corresponding Precaution.	**3**
			Ep(0	No / Irrelevant error or error only.	**0**
6	**DATA PRESENTATION**	**PD**	PD(3	• Realistic organised data of corresponding independent and dependent variables in a suitable table.(I, V) • Appropriate units of independent and dependent variables labelled in appropriate columns. • At least 5 sets of independent and dependent variable values. • Values recorded in expected trend. (Both I and V values increasing or decreasing)	**6**
			PD(2	• Realistic organised data of corresponding independent and dependent variables in a suitable table.(I, V)	**4**

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				• Appropriate units of independent and dependent variables labelled in appropriate columns. • 3–4 sets of independent and dependent variable values.	
			PD(1)	• Realistic organised data of corresponding independent and dependent variables in a suitable table.(I, V) • Appropriate units of independent and dependent variables labelled in appropriate columns. • 1–2 sets of independent and dependent variable values	**2**
			PD(0)	• No set of data	**0**
7	**RECORDING DATA WITH ACCURACY**	**RD**	RD(2)	• At least three (3) dependent variable data recorded within the expected range • At least 3 values of dependent variable data recorded to appropriate decimal places in accordance with the instrument used. • Correct significant figures or decimal places for calculated values. **Note**: **I** must be recorded to **2** decimal places **V** cab be recorded to **0 dp, 1 dp or 2dp** depending on the accuracy of the instrument	**4**
			RD(1)	• 1–2 dependent variable data recorded within expected range. • Dependent variable data has appropriate decimal places in accordance with the instrument used. • Correct significant figures or decimal places for calculated values.	**3**

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			RD(0)	• No / Irrelevant data recorded	**0**
8	**DATA ANALYSIS**	**DA**	DA(3)	• Correct title of the graph • Vertical and horizontal axes with starting points clearly indicated. • Labelling of the axes with quantity and unit. • Suitable and uniform Scales on vertical and horizontal axis. • Correct plotting of at least 5 points. • Line of best fit balancing all the correctly plotted points.	**5**
			DA(2)	• Correct title of the graph • Vertical and horizontal axes with starting points clearly indicated. • Labelling of the axes with quantity and unit. • Suitable and uniform Scales on vertical and horizontal axis. • Correct plotting of 3 - 4 points. • Line of best fit balancing all the correctly plotted points.	**4**
			DA(1)	• Correct title of the graph • Vertical and horizontal axes with starting points clearly indicated. • Labelling of the axes with quantity and unit. • Suitable and uniform Scales on vertical and horizontal axis.	**3**

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				• Correct plotting of 1 - 2 points. • Line of best fit balancing all the correctly plotted points.	
			DA(0)	• No point plotted correctly	**0**
	DATA INTERPRETATION	**DI**	DI(2)	• Substitutes all the four values correctly in the slope formula. • Obtains correct value based on the substituted values. • Correct unit of the value of slope. • Comparison of the value obtained with the range values.	**5**
			DI(1)	• Substitutes 2–3 values correctly in the slope values formula. • Obtains correct value based on the substituted values. • Correct unit of the value of slope. • Comparison of the value obtained with the range values.	**4**
			DI(0)	• Less than 2 values substituted in the slope formula.	**0**
	ADVICE OR DECISION	**CAd**	CAd(1)	• Based on the obtained value and its position in the expected, gives advice or decision for next course of action i.e Conclusion with advice Correctly.	**6**
			CAd(0)	Conclusion without correct advice	**0**

Total Maximum Weight = 52

END

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